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HTA Analytics: A Python Library and Graphical User Interface for Multi-Criteria Decision Analysis in Health Technology Assessment

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Software

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Summary

Health Technology Assessment (HTA) is a formal, multidisciplinary process used to evaluate the clinical, economic, and technical value of medical devices and healthcare technologies (Rogalewicz & Kamenský, 2016). Clinical procurement boards are frequently faced with the challenge of selecting medical systems based on highly heterogeneous, conflicting criteria (e.g., purchase cost, warranty length, software interoperability, and clinical throughput).

`hta-analytics` is a comprehensive Python package accompanied by an intuitive Graphical User Interface (GUI) engineered to run Multi-Criteria Decision Analysis (MCDA) tailored for HTA requirements. The software bridges the gap between advanced decision-science methodologies and clinical administrators, enabling comprehensive analytical pipelines without requiring any programming knowledge.

Statement of Need

While independent mathematical packages for single MCDA methods are available across various open-source ecosystems, existing solutions lack integrated end-to-end pipelines that reflect the empirical workflows of clinical managers. Hospital decision-makers require an environment that supports data ingestion, structural filtering, and mathematical adjustments within a centralized application. Furthermore, determining the robustness of a ranking against variations in criteria weight profiles typically requires complex, custom simulation scripts.

`hta-analytics` addresses these operational gaps by delivering a feature-rich, production-ready framework. The architectural contributions of the library are structured into five core pipelines:

- **Data Ingestion and Standardization**: Supports flexible multi-format data imports (including Excel sheets and CSV datasets) containing complex technical metrics and heterogeneous medical parameters.
- **Condition-Based Knockout Filtering**: Employs customizable logical evaluation engines allowing users to filter alternatives based on strict baseline constraints (e.g., mandatory certifications or minimum clinical capacity thresholds).
- **Criteria Weight Control**: Provides interactive controls to adjust, normalize, and distribute importance weights across competing analytical attributes under mathematical consistency constraints.
- **Algorithmic Variety and Custom Normalization**: Features a wide array of linear and compromise decision models including Simple Additive Weighting (SAW) (MacCrimmon, 1968), TOPSIS (Hwang & Yoon, 1981), and VIKOR (Opricovic & Tzeng, 2004). Scaling discrepancies among disparate indicators are resolved through eight dual-mode benefit/cost normalization transformations evaluated by Vafaei et al. (2016).

- **High-Precision Sensitivity Profiling**: Automates structural validation by incorporating an advanced weight sensitivity profiling engine driven by an interval bisection (binary search) algorithm based on the framework introduced by Millek (2019). The software dynamically calculates stability thresholds and projects weight limits using customized symmetric logarithmic (**SymLog**) visualizations.

By offering both an autonomous GUI application for clinical staff and a documented Python API for reproducible research, `hta-analytics` facilitates sound, auditable, and mathematically transparent decisions in health technology procurement.

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